Informatica® Cloud Application Integration
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Simple RAG Consumption with Databricks Mosaic AI

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Preface

Use *Simple RAG Consumption with Databricks Mosaic AI* to learn how to receive a user query, search for relevant context in a vector database, send the query and context to the Databricks Mosaic AI Large Language Model (LLM), and return a comprehensive response. This guide assumes that you have an understanding of the Databricks Mosaic AI Connector concepts.

CHAPTER 1

Introduction to Simple RAG Consumption with Databricks Mosaic AI recipe

The Simple Retrieval Augmented Generation (RAG) Consumption with Databricks Mosaic AI recipe is based on REST APIs. Use the recipe to receive a user query, search for relevant context in a vector database, send the query and context to the Databricks Mosaic AI Large Language Model (LLM), and return a comprehensive response.

The process begins by receiving a user input prompt. It then converts the query into a vector representation using a pre-trained embedding model, translating the text into a numerical format that captures the semantic meaning.

The process then uses a vectorized query to search for similar vectors within a vector database that contains representations of various contexts. The process retrieves the top K closest matches from the vector database based on similarity scores, representing contexts similar to the user's query.

The process then filters the retrieved contexts to include only those with a similarity score exceeding a specified cutoff parameter, ensuring that only relevant contexts are considered. The process combines these filtered contexts to formulate the final context, which provides additional information to the model.

Subsequently, the process invokes a Large Language Model (LLM) using the original user query and the processed context. The LLM leverages this information to generate a comprehensive response. This response is returned to the user in the same channel.

This approach ensures that the LLM's response is relevant and enriched by the most contextually appropriate information from the vector database.

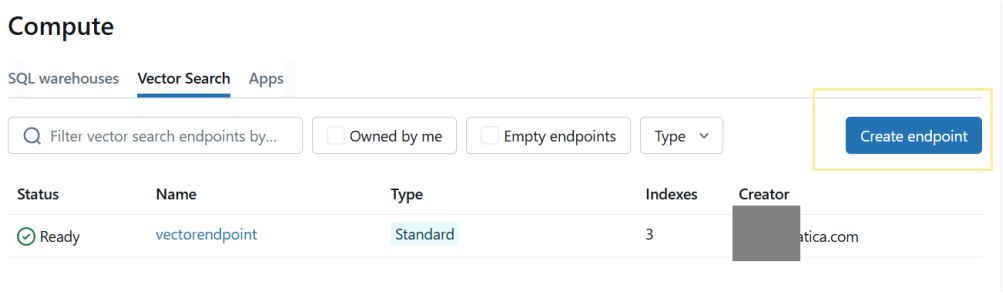
Prerequisites

To use the Simple RAG Consumption with Databricks Mosaic AI recipe, the following prerequisites must be met:

1. **Existing database table**
   You must have a database table available on which the vector index search will be performed.
2. **Enable datafeed**
   Enable the datafeed for this database table to allow continuous data ingestion.
3. **Create a vector endpoint**

In Databricks, create a vector endpoint. A Databricks vector endpoint is a scalable, serverless API hosted on Databricks that enables real-time serving of vector embeddings generated by machine learning models.

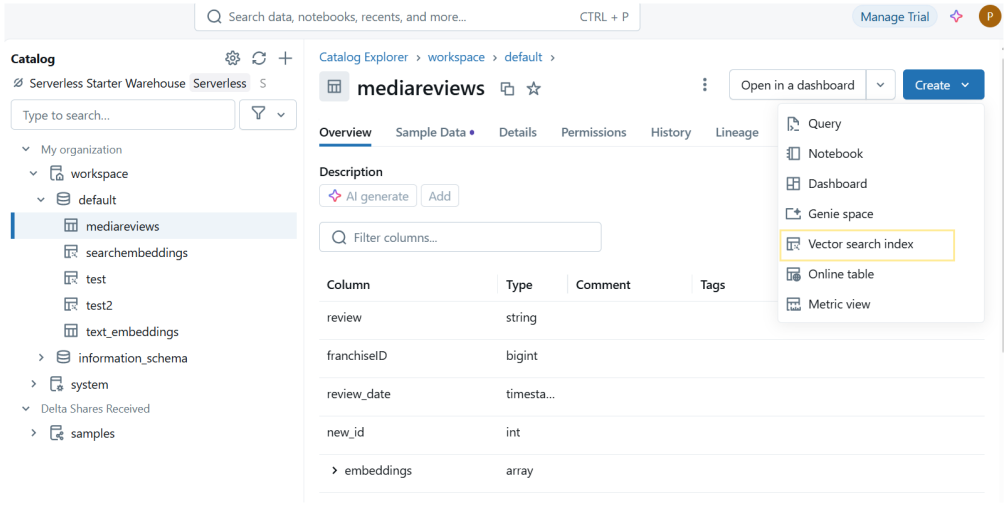
The following image shows the option to create a vector endpoint:



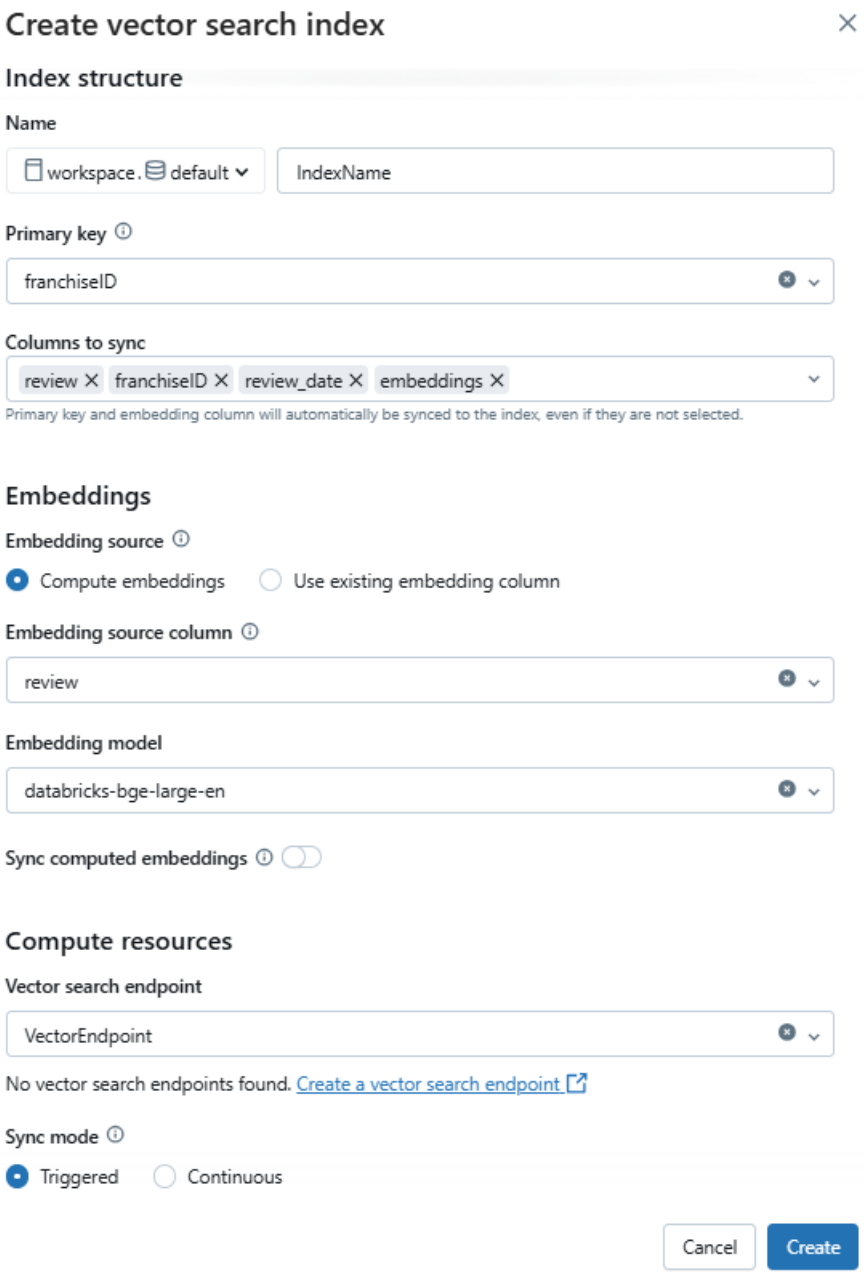
4. **Create a vector search index**
Under the created vector endpoint, create a vector search index for the database table.

- Ensure that the database table has read and write permissions.
- The datafeed must be enabled on the table to support indexing.

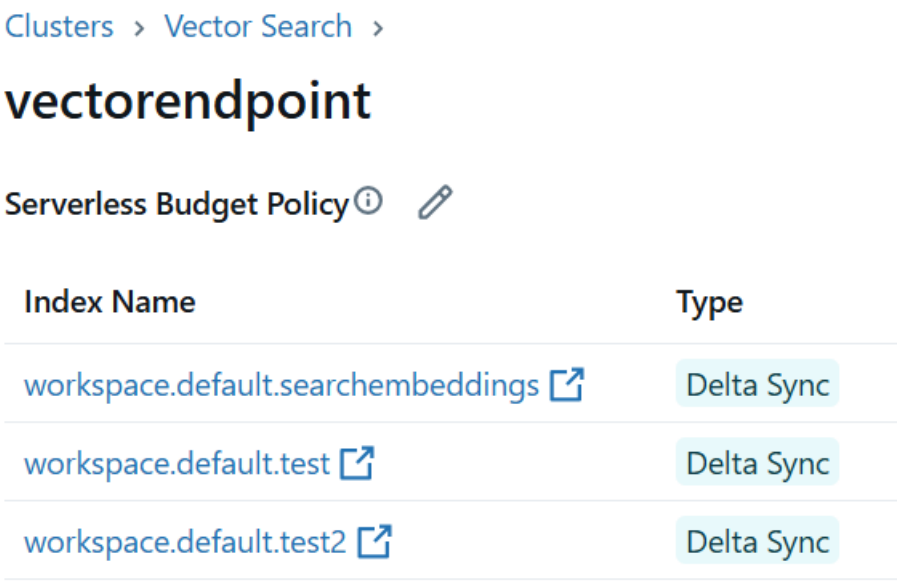
The following image shows the vector search index option:



The following image shows a sample creation of a vector search index:



After creation, the vector endpoint will appear on the Vector Search page as shown in the following image:

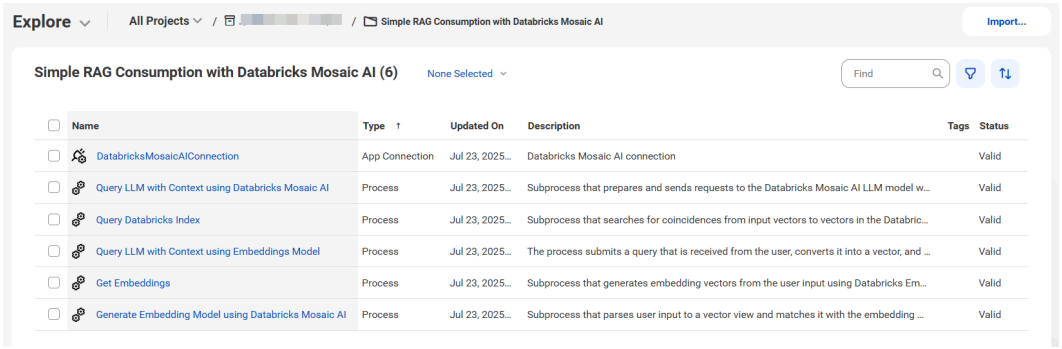

Clusters › Vector Search ›
vectorendpoint
Serverless Budget Policy
Index Name
Type
workspace.default.searchembeddings
Delta Sync
workspace.default.test
Delta Sync
workspace.default.test2
Delta Sync

CHAPTER 2

Simple RAG Consumption with Databricks Mosaic AI recipe contents

The Simple RAG Consumption with Databricks Mosaic AI recipe contains an app connection and processes.

The following image shows the assets that the Simple RAG Consumption with Databricks Mosaic AI recipe package contains:



Simple RAG Consumption with Databricks Mosaic AI recipe assets

The following table lists the assets that the Simple RAG Consumption with Databricks Mosaic AI recipe package contains:

Asset Name	Asset Type	Description
DatabricksMosaicAIConnection	App Connection	Databricks Mosaic AI connection.
Generate Embedding Model using Databricks Mosaic AI	Process	Subprocess that parses user input to a vector view and matches it with the embedding model. It returns the matching context score and metadata.

Asset Name	Asset Type	Description
Query LLM with Context using Databricks Mosaic AI	Process	Subprocess that prepares and sends requests to the Databricks Mosaic AI LLM model with the context.
Get Embeddings	Process	Subprocess that generates embedding vectors from the user input using Databricks Embedding Models.
Query Databricks Index	Process	Subprocess that searches for coincidences from input vectors to vectors in the Databricks Index specified as input parameters and returns a result with metadata.
Query LLM with Context using Embeddings Model	Process	The process submits a query that is received from the user, converts it into a vector, and uses it to search for similar vectors in a database. The top K matches are retrieved, filtered by a cutoff score, and used to form a context. This context including the original query is passed to a Large Language Model (LLM) to generate and return a comprehensive response.

CHAPTER 3

Using the Simple RAG Consumption with Databricks Mosaic AI recipe

To use the Simple RAG Consumption with Databricks Mosaic AI recipe, you must perform the following steps manually:

1. Copy and access the recipe.
2. Configure and publish the Databricks Mosaic AI connection.
3. Publish the processes.
4. Invoke the process.

Step 1. Copy and access the recipe

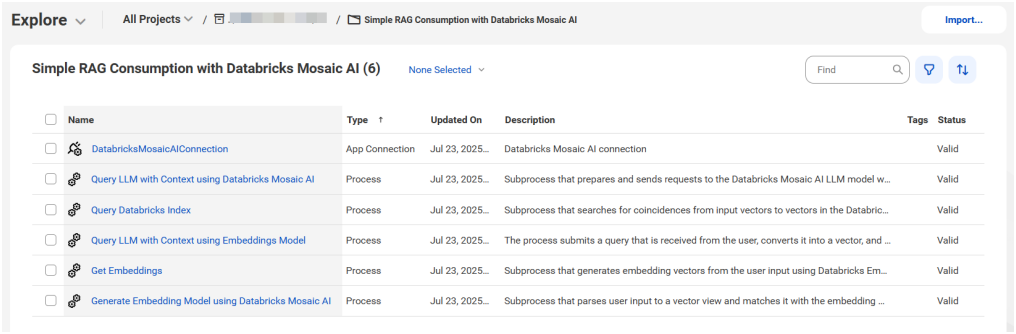
Copy the pre-configured assets in the recipe to a separate project or folder.

1. Open the **Simple RAG Consumption with Databricks Mosaic AI** recipe and click **Use**.
2. Select the location where you want to copy the recipe, and then click **Continue**.
3. In the **Copying the recipe** dialog box, click **OK**.

   It might take some time for the recipe to get copied. You will receive a notification when the recipe is ready for use.
4. After the recipe is copied, click **Explore** to access the recipe content.

5. Navigate to the project or folder where you copied the recipe or enter the recipe name in the **Find** box.

   All the assets in the recipe are displayed as shown in the following image:



Step 2. Configure and publish the Databricks Mosaic AI connection

Configure the host URL and access token to access the Databricks Mosaic AI instance, and then publish the DatabricksMosaicAIConnection connection.

1. Open the **DatabricksMosaicAIConnection** connection.
2. From the **Type** list, select **Databricks Mosaic AI**.
3. From the **Runtime Environment** list, select **Cloud Server or any Secure Agent**.
4. In the **Connection Properties** section, enter values for the following properties:

Property	Description
Host	Host URL for Databricks Mosaic AI.
Bearer_Token	Access token for Databricks Mosaic AI.

5. Save and publish the connection.

Step 3. Publish the processes

To publish the following processes, click **Actions** in the row that contains the processes in the order specified and select **Publish**.

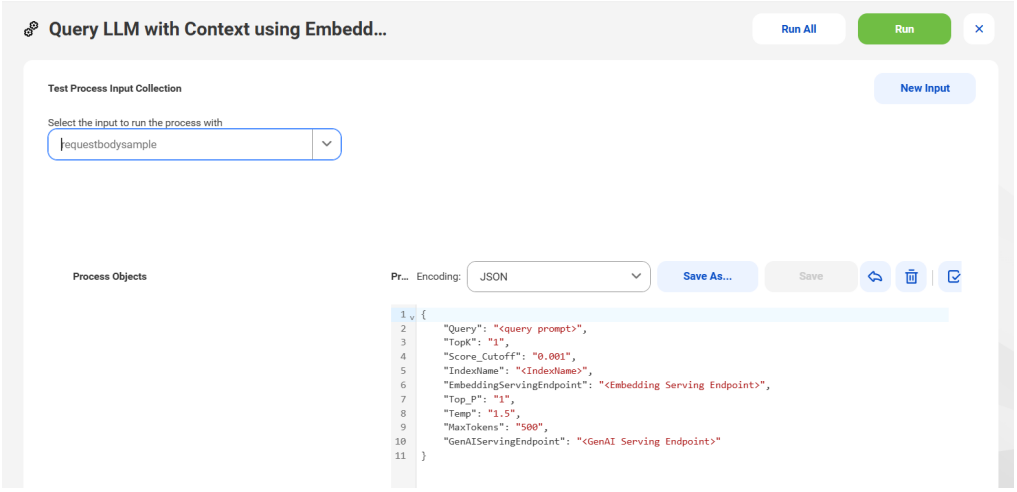
1. Get Embeddings
2. Query Databricks Index
3. Generate Embedding Model using Databricks Mosaic AI
4. Query LLM with Context using Databricks Mosaic AI
5. Query LLM with Context using Embeddings Model

Step 4. Invoke the process

When you invoke the Query LLM with Context using Embeddings Model process, the user sees the answer matching the context that was used in the LLM request.

You can run the process using one of the following options:

1. Open the **Query LLM with Context using Embeddings Model** process and click **Actions** > **Run Using**.
2. Select the input from the **Select the input to run the process with** list and enter the input field and value in the **Process Input** section as shown in the following image:



3. Click **Run**.

You can also invoke the process using a web browser by passing the input parameters, REST or SOAP API endpoints in any API client such as cURL, Postman, SOAP UI, or any programming language.